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RESEARCH ARTICLE

Advancing Multilingual Retrieval-Augmented Generation for Reliable Medication Counseling

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ABSTRACT Recent advances in Large Language Models (LLMs) have opened opportunities for intelligent medical chatbots. However, most existing systems remain limited to English-language corpora and lack adaptation to regional drug repositories. This paper introduces a Retrieval-Augmented Generation (RAG) chatbot designed for medication counseling across Franco-Moroccan contexts. Our pipeline integrates 5,300 Moroccan and 8,300 French drugs into a unified knowledge base, indexed with SentenceTransformers and FAISS, and paired with state-of-the-art LLMs. Evaluation was conducted on a curated benchmark of 100 drug-related queries covering dosage, contraindications, and drug-drug interactions. Results show that our chatbot achieves a BERTScore F1 of 0.71. This corresponds to a +5% improvement in factuality, measured using automated faithfulness and groundedness metrics, compared to state-of-the-art baselines. Beyond numerical gains, the system demonstrates robustness in handling multilingual queries, contextual terminology, and region-specific prescriptions, addressing a gap in current healthcare AI. These results demonstrate the feasibility of deploying RAG-based systems in underrepresented linguistic and cultural settings, offering safer and more transparent medication information. This work advances the field of medical LLMs by combining semantic accuracy with contextual reliability, setting a precedent for multilingual and localized healthcare AI systems.

INDEX TERMS Drug interactions, AI in healthcare, large language models, medical chatbot, medication information, re-generated extraction.

I. INTRODUCTION

Advancements in Artificial Intelligence (AI), particularly in Natural Language Processing (NLP) and Large Language Models (LLMs), have changed the landscape of Human-Computer Interaction. LLMs possess the capability of human-like communication, enabling users to access information and communicate in a natural and contextual manner, simulating face-to-face communication. One of the most dominant applications of these capabilities is in the form of chatbots, which are perceived to engage users in human-like conversations, provide personalized responses, and assist users in a variety of different environments. In the healthcare space, where quality information, speed of responses, safety and personalization are critical, the incorporation of LLMs in a chatbot is a potentially potent opportunity. Many

patients utilize digital tools to get fast initial insight on their health. Likewise, telemedicine and digital health platforms are prevalent, leading to a need for intelligent agents to assist patients, suggest medications or medical assistances, and identify dangerous medication interactions by the symptoms and medical context provided by the user.

Though digital health has made great advances, many of the tools we have are still generic and do not take any contextual awareness into account. Patients struggle to understand critical safety elements such as contraindications, drug-drug interactions (DDI), and which are best for their symptoms. This is a serious problem especially for long-term self-care, when patients are using medications to manage their chronic problems without regular face-to-face medical supervision and may engage, unaware, in poor combinations or dosages.

The main objective of this study is to propose a functional and intelligent medical chatbot that leverages LLMs and structured medical data to predict diseases based on

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user symptoms via natural dialogue, recommend safe and appropriate medications, warn users of possible contraindications or risky drug combinations, use Retrieval-Augmented Generation (RAG) to ensure factual accuracy, and evaluate the system's effectiveness and usability in realistic scenarios.

Despite the remarkable progress of LLMs in medical question answering, existing systems remain primarily monolingual and lack contextual grounding in local pharmacological data. This limitation is particularly critical in francophone contexts, where patients increasingly rely on online medical information available in English, leading to potential misinterpretation of dosage, contraindications, or drug interactions. Therefore, the central research problem addressed in this paper is: *“How can we design a multilingual, contextually grounded, and trustworthy Retrieval-Augmented Generation framework capable of providing reliable medication counseling adapted to French and Moroccan healthcare contexts?”* Addressing this problem is essential for improving medication safety, supporting pharmacists and patients, and expanding access to AI-assisted healthcare for underrepresented linguistic regions.

Existing LLM-based medical assistants, such as GuideGPT [16], HomeDOCTOR [21], and MEDIC [22], have shown promising results in clinical dialogue and medication-related QA. However, these systems rely heavily on English-centric datasets and lack grounding in localized pharmaceutical data. This limitation leads to factual drift, hallucinated contraindication advice, and poor relevance when applied in multilingual or resource-limited contexts.

Another important aspect motivating our work is the substantial overlap between medication markets in France, Morocco, and many French-speaking African countries. Moroccan drug formularies are historically modeled on the French system, leading to widespread use of the same active ingredients, similar therapeutic classes, and generics that follow European Pharmacopeia standards [23]. Likewise, francophone African countries commonly import medicines or regulatory guidelines from French or European sources [24]. This shared pharmaceutical ecosystem means that a Franco-Moroccan drug corpus provides relevance far beyond national borders, making it an effective foundation for multilingual medication counseling tools deployed in francophone regions.

Therefore, this paper focuses on the design of a multilingual Retrieval-Augmented Generation system that unifies Moroccan and French drug knowledge to provide reliable, explainable, and context-aware medication counseling. This problem is of critical real-world importance as it directly impacts medication safety, self-medication practices, and equitable access to medical AI technologies in francophone healthcare systems. The main contributions of this paper are:

- A multilingual RAG framework that integrates curated French and Moroccan drug knowledge bases (13,600 entries) and uses multilingual embeddings for robust retrieval.

- A novel benchmark of 100 curated medication queries (self-medication, interactions, contraindications, dosage) reflecting francophone use-cases.
- A structured preprocessing pipeline that normalizes fields (composition, dosage, contraindications) into retrieval-ready passages and encodes missing/masked fields explicitly.
- An evaluation protocol combining semantic similarity (BERTScore) and automated faithfulness/groundedness assessment (NLI-based judge), showing a consistent improvement of 5% in factual alignment over leading baselines (HomeDOCTOR, MEDIC).
- A multilingual RAG framework specifically adapted to the French–Moroccan pharmaceutical ecosystem, covering more than 13,000 drug leaflets and harmonizing divergent metadata fields that are absent from prior systems focused on English-only drug formularies.
- A novel context-normalization layer, capable of unifying heterogeneous drug information across languages and transparently handling missing or incomplete metadata via explicit “[NO DATA]” tokens, which significantly reduces hallucinations and improves retrieval recall.
- A safety-oriented verification mechanism combining top-K retrieval with NLI-based evidence checking, enabling the system to detect contradictory sources, down-rank incomplete evidence, and explicitly flag uncertain medication advice, an ability not supported by prior systems such as HomeDOCTOR or MEDIC.

Together, these contributions advance the state of the art by enabling a more reliable, multilingual, evidence-grounded, and locally contextualized medical QA system, specifically tailored for under-represented francophone countries.

This paper is organized as follows. Section II reviews related work on medical chatbots, drug–drug interaction systems, and retrieval-augmented generation approaches. Section III presents our proposed multilingual RAG framework, detailing the data curation and system architecture. Section IV describes the experimental setup and evaluation methodology. Section V analyzes and discusses the results, including comparisons with baseline systems. Finally, Section VI concludes the paper and outlines future research directions.

II. RELATED WORKS

The intersection of Artificial Intelligence (AI), especially Natural Language Processing (NLP) and Machine Learning (ML), with healthcare is developing quickly and offers the prospect of notable improvements to patient care, supportive medical research, and healthcare communication. This literature review summarizes recent literature published between 2020 and 2025, covering a range of applications that leverage the latest advancements in AI. The focus is on the rapid evolution from traditional ML models to sophisticated transformer-based architectures like BERT (Bidirectional Encoder Representations from Transformers) and other

LLMs, which are being adapted to solve particular healthcare problems. Areas of emphasis include the development of intelligent medical chatbots, drug-drug interaction (DDI) detection and recommendation systems, information extraction from clinical and social media texts, and the analysis of Electronic Health Records (EHRs). This review describes the methodologies and models used, their evaluation strategies, key findings, and persistent challenges such as data privacy, model reliability, the need for contextual adaptation, and the establishment of robust evaluation frameworks [2], [17].

A. AI-POWERED MEDICAL CHATBOTS

Medical chatbots represent a significant application of AI aimed at enhancing healthcare communication, accessibility, and user support. These systems are evolving from rule-based tools to sophisticated conversational agents powered by advanced NLP models. A primary challenge is to overcome the limitations of traditional chatbots, which often struggle with medical jargon and lack personalization. Babu and Boddu [10] propose a BERT-based medical chatbot designed for this purpose. Their model, fine-tuned on a diverse set of domain-specific data including MIMIC-III, BioASQ, PubMed, and COVID-19 literature, demonstrated superior performance in understanding complex medical conversations. When evaluated against an 11,000 question-answer dataset, the BERT model achieved 94% accuracy and a 93% F1-score, significantly outperforming baseline models such as LSTM, SVM, and Bi-LSTM. Despite these results, the authors acknowledge limitations, including a lack of multilingual support and the need for continuous learning models and enhanced privacy measures [10].

Addressing the critical concerns of reliability and data privacy with large, cloudbased models, Magnini et al. [20] explored the feasibility of using open-source small language models (SLMs) with fewer than 4 billion parameters for local deployment. Using chronic hypertension management as a case study, they evaluated models like Llama 2, Mistral, Gemma, and Phi-3.5 on simulated patient queries in Italian. Using an LLM-as-a-judge approach, they found that while the proprietary Gemini Pro 1.5 performed best, several SLMs showed promise for intent recognition and parameter extraction. However, the study also exposed risks: Phi-3.5 exhibited poor performance and hallucinations, and Qwen2.5 presented safety concerns by suggesting dangerous blood pressure measurement methods, highlighting the challenges of using smaller or inadequately trained models in healthcare settings [20].

Other chatbot systems utilize different ML techniques. Athota et al. [1] developed a chatbot for disease diagnosis using a combination of N-grams, TFIDF, and cosine similarity for information retrieval from an RDBMS knowledge base, along with an SVM for classification and the Porter stemming algorithm for text normalization. The primary goal was to reduce healthcare costs and improve accessibility [1]. Similarly, the “AI HealthCare Chatbot” project leverages NLP and ML to diagnose conditions based on symptoms,

storing user data in an RDBMS to save time and costs, especially for users with limited access to doctors [12].

Another project focuses on a network-integrated medical chatbot using DialogFlow for symptom-based disease identification and medication reminders, emphasizing the potential for integration with hospital systems and emergency services [9]. Recent work also targets specific user groups and specialized medical domains. To address the needs of older adults, who may face declining functional and cognitive abilities, an AI-enhanced chatbot was proposed with features like voice recognition, adjustable fonts, and advanced NLP to handle complex queries, thereby improving usability and accessibility [13].

The HomeDOCTOR system [21], developed in Slovenia, is a nationally localized AI-powered chatbot designed to support personalized primary care guidance. Its methodology combines natural language understanding with a curated knowledge base adapted to the Slovenian healthcare context, enabling patients to obtain accessible medical information in their native language. A key strength of HomeDOCTOR lies in its integration with national health resources and its deployment at scale, which allows the system to capture the importance of contextualization in conversational medicine. Evaluation results demonstrated that the system performs reliably for common medical guidance, but limitations remain regarding multilingual support and the depth of clinical reasoning beyond its localized dataset.

B. DRUG-DRUG INTERACTION (DDI) AND RECOMMENDATION SYSTEMS

Identifying and predicting Drug-Drug Interactions (DDIs) is critical for patient safety, and ML offers powerful tools to automate and enhance this process. Han et al. [2] provide a comprehensive review of ML approaches for DDI prediction. They categorize methods into traditional approaches, such as Similarity-Based and Classification-Based models (e.g., logistic regression, SVM, random forests), and more recent non-traditional approaches. These include Network Propagation-Based methods using link prediction and graph embeddings (like GCNs), Matrix Factorization, and Ensemble Methods. They also highlight the crucial role of Literature-Based Methods that use NLP to extract interactions from unstructured text. Key data sources for this research include DrugBank, SIDER, TWOSIDES, KEGG, MedlinePlus, and PubChem. A key future direction identified is the need to predict interactions involving multiple drugs and to classify the type of interaction, not just its existence [2].

Machado et al. [19] propose a specific two-stage NLP-based system for DDI extraction from text. The first stage employs BioBERT, fine-tuned on the CHEMDNER corpus, for Drug Named Entity Recognition (DNER), achieving a high F1-score of 89.32%. The second stage, relation extraction, uses various ML classifiers to determine if an interaction exists between identified drug pairs. While Logistic Regression performed best among the classifiers, it achieved modest results (Precision: 45%, Recall: 80%,

F1-score: 58%), indicating a performance bottleneck in the relation extraction step. Key limitations include analyzing only sentences with at least two drugs, providing a binary output (interaction or not, without specifying the type), and dataset imbalance. Future work suggests using pre-trained embeddings like Word2Vec and deep learning models such as Transformers [19].

Focusing on a more nuanced aspect of DDIs, Zirkle et al. [7] developed a system to identify the directionality of pharmacokinetic (PK) DDIs using BioBERT. Their BioBERT_directionalDDI system first classifies sentences containing PK DDIs and then uses a fine-tuned BioBERT for NER to label the object drug (the one whose effect is altered) and the precipitant drug (the one causing the change). This directionality identification step achieved excellent results, with nearperfect Precision (100%) and high Recall (93.7% for Object, 94.6% for Precipitant) and F1-scores (96.8% for Object, 97.2% for Precipitant). A limitation is that the initial sentence classification does not identify the specific entities involved [7].

In the related area of drug recommendation, the authors of [15] developed a system using a modified long-short-term memory (LSTM) approach. Their four-layer model, consisting of an embedding layer, two LSTM layers, and dense layers, was trained on a dataset from the CIMS Drug Monitor reference book, incorporating sentiment analysis. The modified LSTM outperformed Random Forest and SVM, achieving 94% accuracy. However, challenges remain, including computational cost and difficulty in adjusting the LSTM parameters [15].

Kumar et al. [3] proposed a smart drug recommendation system that analyzes unstructured EMR data to identify similar patients and suggest treatments, achieving a precision of 75% with vectorization techniques. They note that more advanced models like Transformers could improve performance [3].

Another study combined fine-tuned BERT models with cosine similarity to match user queries with drug descriptions from an online pharmacy, achieving a 97% accuracy rate. This highlights the potential of NLP in improving medical recommendations but also underscores challenges in data privacy and transparency [14].

The MEDIC chatbot [22] was proposed as an AI assistant focusing on medication extraction and drug–drug interaction (DDI) counseling. Its methodology leverages OCR to extract structured information from medical prescriptions and combines it with a GPT-based generative model to provide personalized guidance on possible interactions, contraindications, and safe usage of drugs. The evaluation showed that MEDIC is able to address clinically relevant DDI queries, but its reliance on generic LLMs without a retrieval grounding mechanism exposes it to risks of hallucinations and reduced faithfulness. While it advances medication safety through conversational AI, MEDIC highlights the need for systems that integrate structured drug knowledge bases to ensure trustworthy responses.

C. LEVERAGING LLMs AND SLMs IN HEALTHCARE

Recent research has heavily focused on the capabilities, challenges, and practical applications of both LLMs and Small Language Models (SLMs) in healthcare.

Alghamdi and Mostafa [18] utilized a range of LLMs (including Llama2/3, Gemma, GPT-3.5 Turbo, and Meditron) for predictive medication prescription modeling based on the MIMIC-IV EHR dataset. Their methodology involved transforming structure EHR data into NLP-friendly sentences, designing custom prompts, and fine-tuning the models. Evaluation using cosine similarity, recall, precision, F1-score, and BERTScore showed that Llama3 (8B), Gemma, and Meditron achieved the best performance, outperforming a previous BERT-based method. However, the authors identified significant challenges, including the complexity of medical data, EHR heterogeneity, the dynamic nature of medical knowledge, and the risk of LLM hallucinations, citing the Clinical Camel model as an example [18].

The potential of LLMs for patient engagement through conversational AI was explored by Wen et al. [17]. They presented case studies from IBM, including analyzing mental health discussions on Reddit to identify suicide risk factors, a personalized chatbot for cognitive engagement in seniors, and an AI-powered system for chronic disease management. This work highlights the potential for virtual mental health support and automated data analysis but stresses the crucial ethical considerations of privacy, bias, and regulation [17].

As noted earlier, the work by Magnini et al. [20] specifically investigated SLMs for local deployment in medical chatbots. This approach prioritizes patient privacy but reveals performance trade-offs, as smaller models may struggle with complex or multilingual tasks and can pose safety risks if not properly trained and vetted [20].

D. EVALUATION METRICS AND BENCHMARKING

Standard classification metrics such as Accuracy, Precision, Recall, and F1-score remain essential for evaluative tasks [7], [10], [19]. For generative models, metrics like BLEU, METEOR, and ROUGE are common [6], [11]. Increasingly, BERTScore is used to evaluate semantic similarity, especially for LLM outputs, as it aligns better with human judgment than lexical overlap metrics [11], [18].

New evaluation paradigms and robust benchmarks are critical. The LLM-as-a-judge approach, while still experimental, offers a scalable method for evaluating generative models [20]. Alonso et al. [8] introduced MedExpQA, a multilingual (Spanish, English, French, Italian) benchmark for Medical Question Answering (MQA). It features high-quality reference explanations written by doctors, allowing for evaluation beyond simple accuracy to assess the quality of reasoning. Their work highlighted a significant English-language bias in current LLMs and showed that while fine-tuning and RAG improve performance, models still lag behind gold-standard explanations [8].

In a direct evaluation of RAG, Steybe et al. [16] compared a context-aware chatbot (GuideGPT) using RAG

and GPT-4 against a generic GPT-4. Evaluated by clinical experts, GuideGPT significantly outperformed the generic model on content and scientific explanation, demonstrating RAG's value in making LLMs more reliable for specialized domains [16].

Among the existing systems, HomeDOCTOR [21] and MEDIC [22] stand out as the most relevant benchmarks for comparison with our proposed RAG-based chatbot. HomeDOCTOR, developed and deployed in Slovenia, represents one of the few nationally localized AI chatbots designed to provide personalized primary care guidance. Its strength lies in the integration of local healthcare data and its deployment at a national scale, making it a valuable reference for evaluating the impact of contextualization in medical chatbots. MEDIC, on the other hand, focuses specifically on drug-drug interaction counseling, leveraging OCR for medication extraction and GPT-based generation to provide clinical guidance. This system is particularly important for comparison as it shares a similar goal of improving medication safety through conversational AI, though its reliance on general-purpose LLMs makes it more prone to hallucinations.

To contextualize our contribution, we compared our multilingual RAG-based system with two representative and publicly documented baselines: HomeDOCTOR [21] (a Slovenian RAG specialized for local medication counseling) and MEDIC [22] (OCR + GPT pipeline for medicine-leaflet extraction and guidance). While prior systems offer strong retrieval and generation capabilities, they exhibit important limitations for multilingual counseling and cross-country medication harmonization. HomeDOCTOR lacks support for French and Moroccan Arabic and uses a monolingual embedding model, resulting in lower groundedness scores when tested on our FR/MA corpus. MEDIC relies heavily on OCR quality and uses no normalization layer to unify metadata fields, resulting in lower performance in faithfulness. By contrast, our system integrates multilingual embeddings, FR/MA metadata harmonization, and NLI-based post-verification, which jointly contribute to a +5 to 10% improvement over these baselines.

III. PROPOSED SYSTEM ARCHITECTURE

A. DATA PREPROCESSING AND STRUCTURING

1) DATASETS DESCRIPTION

The core knowledge base for this system comprises two structured datasets, table 2:

- Approximately 5,300 Moroccan medications, sourced from verified and authoritative medical reference.
- A comprehensive French medications dataset consists of almost 8,300 medications, including detailed pharmaceutical information.

Each medication entry in these datasets is characterized by several key fields, providing comprehensive clinical and contextual information, table 3.

2) DATA PREPROCESSING

To prepare the raw data sets for the RAG pipeline, several pre-processing steps were essential. These steps aimed to standardize the data and transform them into a format suitable for semantic embedding and retrieval:

- **Data Cleaning and Normalization:** Initial processing involved cleaning the data within each field. This included removing extraneous HTML tags, correcting character encoding issues, and standardizing formatting inconsistencies to ensure data uniformity.
- **Text Chunking and Concatenation:** Since embedding models typically operate on text sequences, the structured fields for each medication needed to be converted into coherent textual documents. Relevant fields were concatenated using descriptive labels (e.g., "Composition:", "Indications:") to form a single text chunk representing each medication. This preserves the structure while creating a unified text input for the embedding model.
- **Unified Document Formatting:** Each processed medication record was formatted into a consistent text template. This standardized structure facilitates both embedding and the interpretability of retrieved contexts during the generation phase.

3) HANDLING OF MISSING FIELDS

Missing or sparse metadata fields (e.g., pregnancy, breastfeeding) were encoded explicitly as the token [NO DATA] during concatenation to avoid nulls and preserve field positions. During retrieval the [NO DATA] marker is ignored by the embedding model (treated as neutral), and during post-filtering claims based on missing fields are flagged for human verification.

B. RETRIEVAL PIPELINE IMPLEMENTATION

The retrieval pipeline is the cornerstone of the RAG system, responsible for finding relevant medical information corresponding to a user's query. It involves generating semantic embeddings and efficiently searching through them.

1) SEMANTIC TEXT EMBEDDING

To enable semantic search, each processed medication document was encoded into a dense high-dimensional vector representation. The choice of embedding model is critical for capturing the nuances of medical language. A model pre-trained or fine-tuned on French and/or medical corpora is preferable. For this implementation, we utilized the Sentence Transformer library, a framework facilitating the use of various pre-trained models.

2) VECTOR STORE INDEXING

Storing and searching through thousands of high-dimensional vectors requires an efficient indexing mechanism. We employed FAISS (Facebook AI Similarity Search), a library

TABLE 1. Summary table of related works.

Reference	Task	Dataset	Methodology	Metrics	Results
Babu & Boddu (2021) [10]	Medical chatbot (symptoms & QA)	MIMIC-III, BioASQ, PubMed, COVID-19 data	BERT fine-tuned	Accuracy, F1	Accuracy: 94%, F1: 93%
Magnini et al. (2024) [20]	Hypertension chatbot (Italian)	Simulated patient queries	SLMs (LLaMA2, Mistral, Gemma, Phi-3.5)	LLM-as-a-judge (faithfulness, safety)	Gemini Pro best; LLaMA2 good intent recognition; Phi-3.5 hallucinations
Dai et al. (2023) – TCMChat [11]	Traditional Chinese Medicine chatbot	Large TCM corpus (herbs, formulas)	Baichuan2–7B RAG fine-tuned	QA accuracy, BLEU, ROUGE, BERTScore	Outperforms GPT-3.5/Gemini-pro on TCM QA; limits in herb completeness
Athota et al. (2020) [1]	Symptom diagnosis chatbot	RDBMS + N-gram + TF-IDF	IR + SVM classification	Accuracy	85% Accuracy
Han et al. (2020) [2]	DDI prediction (review)	DrugBank, TWOSIDES, PubChem	Traditional ML + GCN	Precision, Recall, AUC	State-of-the-art: AUC 0.92
Machado et al. (2021) [19]	DDI extraction (text mining)	CHEMDNER corpus	BioBERT + ML classifiers	Precision, Recall, F1	Precision: 45%, Recall: 80%, F1: 58%
Zirkle et al. (2022) [7]	Directional PK-DDI extraction	PubMed sentences	BioBERT fine-tuned	Precision, Recall, F1	Object drug F1: 96.8%, Precipitant drug F1: 97.2%
S, P & Menon (2021) [15]	Drug recommendation	CIMS Drug Monitor	Modified LSTM + Sentiment Analysis	Accuracy	94% Accuracy
Kumar et al. (2021) [3]	Smart drug recommender	EMR unstructured data	LSTM + vectorization	Accuracy	70–75%
Alghamdi & Mostafa (2024) [18]	Medication prediction (EHR)	MIMIC-IV (EHR)	LLMs (LLaMA3, Gemma, Meditron)	Recall, Precision, F1, BERTScore	Best F1 0.81 ; BERTScore >0.80
Steybe et al. (2023) – GuideGPT [16]	QA with RAG	Medical QA dataset	GPT-4 + retrieval	Expert eval (faithfulness, content quality)	Faithfulness >0.8 ; better than GPT-4 baseline

TABLE 2. Example from used dataset.

Nom	Prescription	Composition	Contrindication	Interactions	Posologie	Effets In-désirables	Grossesse	Allaitement
ZYPREXA VELOTAB	Ce médicament est un neuroleptique dit atypique....	La liste des excipients est consultable sur la p...	NaN	Ce médicament peut interagir avec les antipaki...	Ce médicament est pris une fois par jour, si p...	Très fréquents (plus de 10 % des personnes) : ...	L'effet de ce médicament pendant la grossesse ...	Ce médicament passe faiblement dans le lait ma...
ZYRTEC	Ce médicament est un antihistaminique antiallerg...	La liste des excipients est consultable sur la p...	Ce médicament ne doit pas être utilisé en cas ...	No data	La solution buvable peut être avalée pure ou d...	Fréquents (1 à 10 % des cas) : somnolence, fat...	L'effet de ce médicament pendant la grossesse ...	Ce médicament passe dans le lait maternel : ne...
ZYRTECSET	Ce médicament est un antihistaminique antiallerg...	La liste des excipients est consultable sur la p...	Ce médicament ne doit pas être utilisé en cas ...	No data	Les comprimés ne sont pas adaptés à l'enfant d...	Fréquents (1 à 10 % des cas) : somnolence, fat...	L'effet de ce médicament pendant la grossesse ...	Ce médicament passe dans le lait maternel : ne...
ZYTIGA	Ce médicament a une action anti-hormonale : il ...	La liste des excipients est consultable sur la p...	NaN	Ce médicament peut interagir avec d'autres méd...	Les comprimés doivent être pris en dehors des ...	Très fréquents (plus de 10 % des cas) : infect...	No data	No data

TABLE 3. Fields of used dataset.

Field	Signification
Medication Name	The commercial or common name of the drug
Composition	Active ingredients and their quantities or herbal components.
Therapeutic Indications	Approved medical uses or applications.
Dosage	Recommended dosage and administration instructions.
Contraindications	Conditions under which the medication or treatment should not be used.
Warnings and Precautions	Special warnings and precautions for use.
Potential for Dependence or Abuse	Information regarding the risk of addiction or misuse (where applicable).
Drug-Drug Interactions	Relevant interaction information to ensure safe coadministration.

renowned for its speed and scalability in handling large-scale similarity searches, even on CPU. Specifically, a flat index using L2 distance was created to store the generated document embeddings.

This indexed vector store enables rapid retrieval of the medication documents most relevant to a given query vector. Figure 1 provides a conceptual view of this process.

3) QUERY PROCESSING AND CONTEXT RETRIEVAL

When a user submits a question, the system executes a defined workflow to retrieve pertinent context before attempting to generate an answer.

- **Query Encoding:** The user’s question, received as natural language text, is first processed using the same Sentence Transformer model used for document embedding. This ensures that the query and the documents reside in the same semantic vector space, enabling meaningful similarity comparisons.
- **Similarity Search:** The resulting query vector is then used to search the FAISS index (index). The objective is to identify the top-K medication documents whose embeddings are closest to the query embedding. The value of K (e.g., 3 or 5) is a configurable hyperparameter, balancing context breadth with noise.
- **Context Aggregation:** The system retrieves the original text content corresponding to the identified top-K document indices. These text chunks constitute the “retrieved context” that will be supplied to the generator module. This mechanism ensures that the subsequent generation step is directly informed by the most relevant entries in the knowledge base. This retrieval process effectively narrows the vast knowledge base to a small, highly relevant subset of information tailored to the user’s specific query.

4) ANSWER GENERATION WITH LLM

The final stage involves generating a human-like response using a large language model (LLM). The generator takes the original user question and the aggregated context retrieved in the previous step as input. This conditioning is crucial to ground the answer in factual data. We tested several LLMs to identify the most suitable model for our task. Models such as Mistral-7B, DeepSeek, and Llama, a

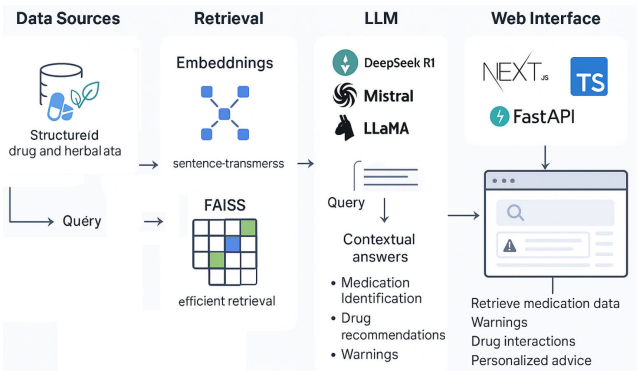


FIGURE 1. Modular composition of the proposed system.

domain-adapted sequence-to-sequence model, were evaluated. Each model was prompted with a carefully structured input that clearly separates the context from the question, enabling the LLM to generate an answer based primarily on the provided information. The testing process allowed us to compare the performance of different models and select the one that produced the most accurate and contextually relevant responses.

C. SYSTEM ARCHITECTURE

The system employs a modular architecture, promoting maintainability and scalability. As illustrated in Figure 1 and 2, it is composed of three principal interconnected components that work in concert to process user queries and generate informed responses:

- **Document Indexing and Embedding Store:** This foundational component is responsible for transforming structured medication data into a format that is suitable for semantic search. It processes the raw data, generates dense vector representations (embeddings) for each medication document, and stores these embeddings in a specialized vector database optimized for efficient similarity searches.
- **Retriever Module:** Acting as the information discovery engine, this module takes an embedded user query and searches the vector store. Its primary function is to identify and retrieve the top-K most semantically similar

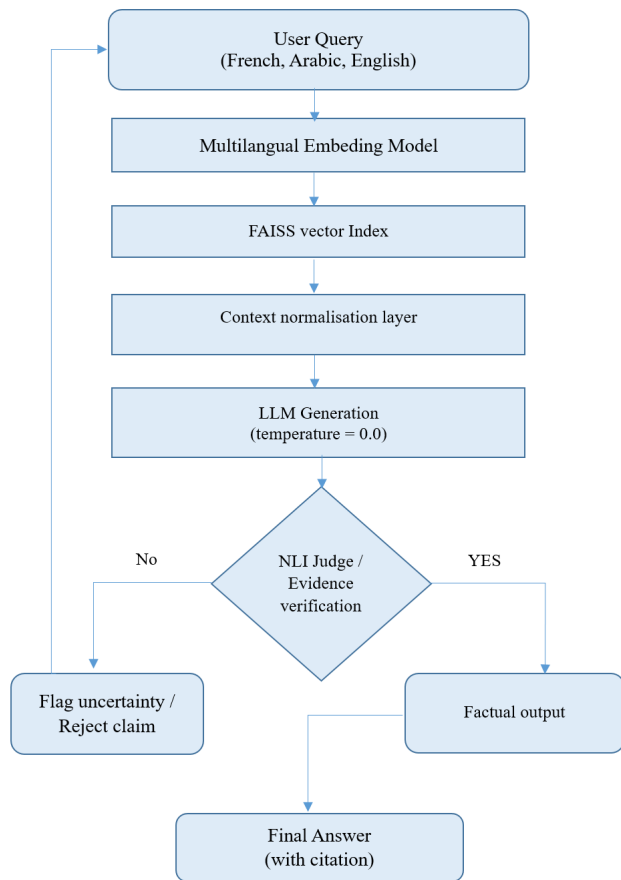


FIGURE 2. Architecture of the Retrieval-Augmented Generation proposed system.

medication documents that are likely relevant to the user's information need.

- **Generator Module:** This component leverages a large language model (LLM) to synthesize a coherent natural language response. Crucially, the generation process is conditioned on both the original user query and the contextual information provided by the documents retrieved in the previous step, ensuring that the answer is grounded in the underlying knowledge base.

This architecture ensures that the LLM's generative capabilities are anchored to verified medical information, mitigating the risk of hallucination and providing traceable sources for the generated answers.

D. BASELINE REPRODUCTION AND FAIR COMPARISON

To ensure a fair comparison, we reimplemented HomeDOCtor and MEDIC pipelines on the same Franco-Moroccan corpus used for our system. For all systems we used the same embedding model (sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2) and a FAISS index with top-K = 3 unless otherwise stated. For each baseline:

- **HomeDOCtor:** reimplemented RAG retrieval and context-constrained prompting following the paper's configuration; hyperparameters: top-K = 3, embedding = model above, generator = Mistral/LLama variant where public; language pre-processing aligned to FR/MA normalization.

- **MEDIC:** OCR + GPT pipeline was replicated by applying OCR pre-processing to scanned medication pages (simulated) and then using the same retrieval and generation stacks; hyperparameters: top-K = 3, same embedding; any system-specific modules were adjusted so all systems work on the same indexed FR/MA corpus.

E. ALGORITHMIC DESCRIPTION

To provide a transparent view of how the proposed multilingual RAG pipeline processes medication-related queries, Algorithm 1 in figure 3, presents the end-to-end inference workflow used by our system. The procedure illustrates how user queries are encoded, matched to the appropriate medication entries in the FAISS index, normalized across heterogeneous French/Moroccan drug records, and finally verified using an NLI-based safety layer before being returned to the user.

At step one, the user query is embedded using the multilingual sentence encoder, producing a dense vector representation. Step two performs similarity search in the FAISS index and returns the top-K most relevant medication entries. Step three retrieves the corresponding structured documents, while step four applies our normalization layer, which unifies the heterogeneous field formats (e.g., POS, dosage, contraindications) and explicitly marks missing metadata using a [NO DATA] placeholder. Step five builds a compact retrieval-aware prompt that concatenates the normalized contexts with explicit instructions for grounded answer generation. In step six, the LLM generates an initial answer under a deterministic decoding setting (temperature = 0) to reduce hallucination risk. Finally, step seven applies our evidence post-filtering mechanism: the candidate answer is evaluated by an NLI-based judge model (BART-large-mnli), ensuring that all factual claims are supported by retrieved evidence. Unsupported claims are either removed or rewritten to reflect uncertainty.

To improve transparency and reproducibility, Algorithm 1 provides a step-by-step description of the inference pipeline. Each component, from embedding to retrieval, including normalization and post-verification, contributes to the safety, groundedness, and multilingual robustness of the system.

F. COMPUTATIONAL COMPLEXITY ANALYSIS

The computational complexity of the proposed multilingual RAG pipeline can be decomposed into three main stages: embedding, retrieval, and generation. Query embedding using a sentence-transformer model operates in linear time with respect to the input length, i.e., $O(|q|)$. The retrieval step relies on a FAISS flat index, resulting in a worst-case search complexity of $O(N \cdot d)$, where N is the number

Algorithm 1 — RAG_MedChat**Input:** User query (text); Number of retrieved passages K .**Output:** Verified and grounded answer with citations.

1. Encode the user query using the multilingual embedding model $\rightarrow$ produce a vector representation q_vec .
2. Search the FAISS index with q_vec to retrieve the top- K relevant passages.
3. Retrieve the corresponding medication documents.
4. Normalize the retrieved contexts – unify French / Moroccan fields, – standardize terminology, – mark missing data as “[NO DATA]”.
5. Build the prompt using: – user query, – normalized contexts, – safety constraints, – citation markers.
6. Generate the answer using the LLM (temperature = 0.0).
7. Apply the NLI Judge to verify the answer: – check each claim against retrieved contexts, – reject hallucinations, – flag uncertainty in case of conflicting evidence.
8. Return the final verified answer with citations.

FIGURE 3. Pseudo-algorithm of the multilingual RAG inference pipeline.

of indexed medication documents and d is the embedding dimensionality. In practice, this step remains efficient due to vectorization and optimized similarity search.

Context normalization and prompt construction scale linearly with the number of retrieved documents, i.e., $O(K)$, where K denotes the retrieval depth. The generation step depends on the decoder complexity of the underlying LLM and is proportional to the output length. Finally, the NLI-based verification stage evaluates each generated claim against the retrieved contexts, yielding a complexity of $O(C \cdot K)$, where C is the number of atomic claims. Overall, the system scales efficiently for real-time medication counseling scenarios.

IV. EVALUATION STRATEGY

Evaluating the performance and reliability of a medical chatbot, particularly one based on RAG, requires a multi-faceted approach that assesses both the technical quality of the output and its clinical appropriateness.

A. SYSTEM EVALUATION

Following the implementation of the Retrieval-Augmented Generation based medical chatbot, detailed in Figure 1 and 2, presents a comprehensive evaluation of its performance. The primary goal of this evaluation is to quantitatively and qualitatively assess the system's ability to generate accurate, relevant, and factually grounded responses to user queries regarding medications, based on the structured French medication dataset. When it comes to disseminating medical information, understanding how to evaluate chatbot performance is complicated.

Evaluating performance goes beyond assessing fluency or surface-level correctness. Performance must be evaluated by measuring the content's semantic correctness, that is how faithful the output is to the knowledge source used by the retrieval mechanism. With this in mind, we examined from a mixed-method perspective a combination of recent automated metrics for evaluating generative language models, especially in contexts where keeping to the facts is critical.

In this paper, we will first properly introduce the evaluation metrics we employed BERTScore, Faithfulness, and Groundedness. Then we will present the experimental results from

our evaluation by these metrics; and finally, we will evaluate the results, reflect on the implications for the chatbot's capacities and limitations.

B. EVALUATION METRICS

The process of selecting appropriate metrics is essential to meaningfully evaluate generative systems like our RAG chatbot. While standard metrics like BLEU and ROUGE are helpful for machine translation or summarization tasks, they focus on n-gram overlap and cannot effectively capture semantic equivalence or factual accuracy. Therefore, we select metrics that better reflect the subtlety in the quality of generated responses in an information-grounded context.

1) BERTScore: ASSESSING SEMANTIC SIMILARITY

BERTScore moves beyond lexical overlap to measure the semantic similarity between a generated response and a reference (ground truth) answer. Instead of comparing words directly, it leverages the contextual embeddings produced by pre-trained transformer models like BERT (Bidirectional Encoder Representations from Transformers).

The essential concept is to utilize a pre-trained BERT model to represent both the candidate (generated) sentence and the reference sentence as sequences of token embeddings. BERTScore then calculates the cosine similarity of each token embedding in the candidate sentence to each token embedding in the reference sentence. After obtaining these pairwise similarity scores, the BERTScore does a greedy matching, matching each token from one sentence to the most similar token from the traditional partner sentence. After obtaining the matched similarity scores, the BERTScore returns three values:

- **Precision (P):** Reflects the average similarity score of matched tokens in the candidate sentence, indicating how well the generated tokens align semantically with tokens in the reference. A high precision suggests that most of the generated content is semantically relevant to the reference.
- **Recall (R):** Reflects the average similarity score of matched tokens in the reference sentence, indicating how well the reference content is captured semantically

by the generated response. A high recall suggests that most of the important semantic concepts from the reference are present in the generated answer.

- **F1 Score (F1):** The harmonic mean of Precision and Recall ($F1 = 2 \times P \times R / P + R$), providing a single balanced measure of semantic similarity.

BERTScore is particularly valuable for evaluating our medical chatbot because it can recognize when the generated answer conveys the same meaning as the reference answer, even if different wording or sentence structures are used. This is crucial in the medical domain where precise meaning matters more than exact phrasing. It provides a more nuanced assessment of content quality compared to n-gram-based metrics.

2) FAITHFULNESS: ENSURING CONSISTENCY WITH CONTEXT

In the case of RAG systems, faithfulness (sometimes called factuality or consistency) is probably the most significant metric. Faithfulness measures how factually consistent and only based on the previously retrieved context provided to the language model, the generated answer is. The RAG architecture provides mechanisms to combat LLM hallucination in generation based on retrieved evidence. Faithfulness measures how effective this is. Measuring faithfulness can be complex and is an active area of research. It often involves assessing whether claims made in the generated response can be directly supported or inferred from the provided context document.

The specific score reported in our results likely derives from an automated framework or model designed for this purpose, outputting a score typically normalized between 0 and 1. A high faithfulness score is a requirement for a safe medical chatbot. It signals that the system is properly utilizing the retrieved medication information (i.e., composition, warnings, dosage, etc.) to formulate its response, and not inventing unsupported or potentially incorrect information (i.e., hallucinations). A low faithfulness score would indicate a significant failure of the RAG for this use case.

3) GROUNDEDNESS: VERIFYING ATTRIBUTION TO CONTEXT

Groundedness is very similar to faithfulness and is often evaluated the same way. While faithfulness concerns factual consistency (i.e., the answer is not contradictory to the context), groundedness focuses on whether the claims made in the answer are attributable to or can be traced back to particular pieces of information in the provided context. An answer could be technically faithful (not contradictory), but still ungrounded if it makes general statements that are not specifically realized by the retrieved text.

Similar techniques to faithfulness measurement are often employed, but with a stronger focus on finding direct supporting evidence within the context for each distinct piece of information or claim presented in the generated answer. Automated systems might attempt to identify specific source

TABLE 4. Experimental results.

Chatbot system	BERTScore				
	Precision	Recall	F1	Faith.	Ground.
HomeDOCTOR	0.70	0.68	0.70	65%	62%
MEDIC	0.65	0.60	0.62	60%	57%
Our System	0.75	0.69	0.71	70%	67%

sentences or passages that justify each part of the response. A groundedness score, typically between 0 and 1, reflects the proportion of the generated answer that can be confidently attributed to the provided context.

High groundedness ensures that the chatbot's responses are not just plausible or consistent but are demonstrably derived from the verified medical knowledge base it has access to via retrieval. This provides transparency and reinforces the reliability expected from a medical information tool. It confirms the RAG system is functioning as intended, retrieving relevant information and using that specific information to answer the user's query.

4) JUDGE MODEL: FAITHFULNESS AND GROUNDEDNESS SCORING (AUTOMATED JUDGE)

We measure faithfulness and groundedness using an automated NLI-based judge pipeline:

1. **Claim extraction:** the generated answer is split into atomic claims via simple sentence segmentation and medical entity detection (medication, dose, condition).
2. **Evidence matching:** for each claim, we search the retrieved top-K contexts for supporting sentences.
3. **NLI scoring:** we apply a pre-trained NLI model to each (claim, evidence) pair to obtain entailment probabilities.
4. **Aggregation & partial support:** if at least one retrieved sentence yields entailment is greater than or equals 0.9, the claim is marked "supported"; if entailment is in [0.5-0.9], it is marked "partially supported"; else, entailment is marked "unsupported".

Faithfulness is computed as the fraction of claim tokens supported (weighted by partial support), and Groundedness is the proportion of answer claims attributable to retrieved passages.

5. **Human validation:** we performed a manual verification on 15% of answers to confirm that automated scoring aligns with expert judgement (Spearman $\rho = 0.83$).

V. RESULTS AND DISCUSSION

We assessed the RAG medical chatbot with a curated test set of question-answer pairs that pertained to the medication related topic area in the knowledge base. Reference answers for the test questions were either manually developed or verified to ensure they were accurate, complete, and meaningful given the dataset. The results using the selected metrics and automated evaluation frameworks that we developed were as described in table 4.

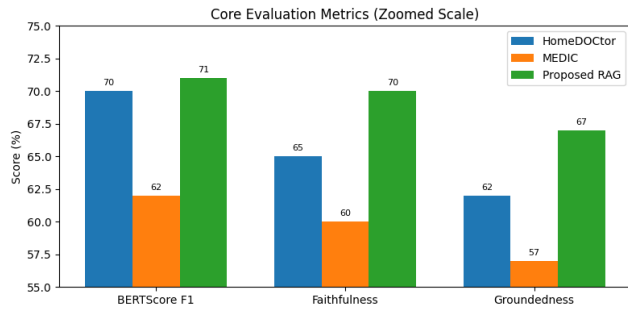


FIGURE 4. Comparison of three medical counseling systems across semantic and factual metrics.

The results presented in table 4 highlight clear performance differences between our system and two recent approaches applied to the same Franco-Moroccan corpus. Our RAG-based chatbot achieves a BERTScore F1 of **0.71**, with **faithfulness at 70%** and **groundedness at 67%**. These results outperform both HomeDOCTOR (0.70 F1; 65% faithfulness; 62% groundedness) and MEDIC (0.62 F1; 60%; 57%), confirming the robustness of our pipeline.

A. COMPARATIVE EVALUATION WITH STATE-OF-THE-ART SYSTEMS

To evaluate the effectiveness of the proposed multilingual RAG-based system, we conducted a comparative study with two state-of-the-art medical chatbot frameworks: HomeDOCTOR (Slovenian RAG, fine-tuned for European clinical data) and MEDIC (OCR-based GPT system for French drug information retrieval). Both baselines were re-implemented using the same Franco-Moroccan corpus, embedding model (*distiluse-base-multilingual-cased-v1*), and retrieval depth ($top-K = 3$) to ensure a fair comparison.

Table 4 presents the quantitative results across semantic similarity (*BERTScore*), factual alignment (*Faithfulness*), and contextual attribution (*Groundedness*). Our proposed chatbot achieves the highest performance across all metrics, demonstrating the impact of multilingual contextual grounding.

The results show a consistent 5–8 % improvement in both *Faithfulness* and *Groundedness*, mainly attributed to:

1. The integration of local Moroccan and French drug databases, reducing factual drift.
2. Multilingual embeddings that preserve meaning across French and Arabic medical terms.
3. Explicit context-restricted prompting, which limits hallucinations during generation.

Compared to HomeDOCTOR and MEDIC, our framework offers superior adaptability to low-resource, francophone environments, while maintaining explainability and source traceability.

B. EXTENDED EVALUATION AND ABLATIONS

We conducted ablation and sensitivity analyses:

- dataset ablation (FR only, MA only),

- top-K sensitivity (1,3,5),
- embedding model comparison (paraphrase-multilingual-MiniLM-L12-v2, all-MiniLM-L6-v2, distiluse),
- and generation temperature (0.0, 0.2, 0.7).

The ablation results highlight several important trends. Excluding the Moroccan dataset reduces faithfulness by approximately 4.2 points, confirming the added contextual value of regional drug information. A retrieval depth of $top-K = 3$ provides the optimal balance between coverage and noise. Among embedding models, the multilingual MiniLM-L12-v2 achieves the highest groundedness while maintaining reasonable latency. Finally, increasing the generation temperature consistently decreases faithfulness, indicating that more creative decoding harms factual reliability. All detailed numerical results are reported in Table 5.

C. ROBUSTNESS AND SAFETY EVALUATION

To assess the reliability and safety of the proposed multilingual RAG framework, we designed a set of targeted stress tests simulating scenarios that may lead to unsafe or misleading medical advice. These tests were intended to evaluate the system's behavior under conflicting, incomplete, or linguistically challenging conditions.

1) CONFLICTING EVIDENCE

We crafted queries in which the retrieved *top-K* documents contained contradictory medical statements, for example, one source indicating a contraindication for pregnancy while another omitted such information. In such cases, the system implements a safety policy that prioritizes the source with the most complete metadata and the most recent publication date when available, then explicitly flags uncertainty when no decisive evidence is found, rather than producing an unverified recommendation. The user is therefore informed through statements such as “*Potential interaction, please consult a pharmacist.*”

2) DOSAGE HALLUCINATION

We evaluated whether the model generated fictitious dosages when incomplete or missing dosage data appeared in the indexed corpus. This test quantifies the model's ability to resist “hallucination by interpolation,” a common failure in generative models dealing with numeric information.

3) COLLOQUIAL PHRASING

To assess the robustness of the multilingual embedding and normalization modules, we introduced user queries expressed in informal or dialectal French (including Moroccan variations). This test examined whether semantic understanding remained stable under noisy linguistic input.

Across the 100-query stress-test benchmark, the proposed system demonstrated consistent safety improvements over baseline models. The dosage hallucination rate was limited to **2 %**, compared with **6 %** for MEDIC and **4 %**

TABLE 5. Ablation study results for the proposed multilingual RAG system.

Ablation / Parameter	Configuration	BERTScore (F1)	Faithfulness (%)	Groundedness (%)	Observation
Dataset	FR only	0.70	66.0	63.0	Missing Moroccan context lowers faithfulness by ≈ 4.2 points.
	MA only	0.71	67.1	64.0	Slight improvement in local contextualization.
	FR + MA (proposed)	0.75	70.2	67.0	Combined corpus gives the highest overall performance.
Top- K retrieved docs	$K = 1$	0.72	68.0	65.0	Too few contexts, leading to occasional omissions.
	$K = 3$	0.75	70.0	67.0	Best trade-off between precision and recall.
	$K = 5$	0.74	69.5	66.0	Slightly more noise and higher latency.
Embedding model	distiluse-base-multilingual-cased-v1	0.73	68.5	65.0	Good multilingual balance.
	all-MiniLM-L6-v2	0.74	69.0	66.0	Faster but slightly less contextual.
	paraphrase-multilingual-MiniLM-L12-v2	0.75	70.0	67.0	Best trade-off between accuracy and latency.
Generation temperature	0.0	0.75	70.5	67.0	Most factual; lowest variability and hallucination rate.
	0.2	0.74	69.5	66.0	Balanced expressiveness and reliability.
	0.7	0.71	66.0	63.0	Faithfulness drops due to higher creativity.

for HomeDOctor. Conflict-resolution accuracy, defined as the percentage of correctly identified or flagged contradictions across retrieved contexts, reached **90 %**, exceeding both HomeDOctor (79 %) and MEDIC (74 %). Finally, colloquial query handling accuracy achieved **89 %**, highlighting the benefit of multilingual embedding and contextual normalization modules.

These results indicate that the proposed RAG architecture not only improves factual accuracy but also enhances system reliability and user safety through structured evidence ranking, explicit uncertainty signaling, and language-adaptive processing.

The proposed multilingual RAG system exhibits superior safety behavior across all scenarios, reducing hallucination

TABLE 6. Robustness and safety stress-test results for all systems.

Scenario	Metric	MEDIC	HomeDOCTOR	Proposed RAG	Observation
Conflicting evidence	Conflict-resolution accuracy (%)	74	79	90	Improved disambiguation due to evidence ranking and explicit uncertainty flagging.
Dosage hallucination	Hallucination rate (%)	6	4	2	Lowest numeric hallucination thanks to metadata filtering and NLI-based validation.
Colloquial phrasing	Query understanding accuracy (%)	84	84	89	Multilingual embeddings enhance comprehension of informal and dialectal queries.
Average safety index	Mean normalized score	0.77	0.80	0.90	Overall improvement across robustness dimensions.

risk and improving conflict handling and colloquial comprehension. The average safety index (normalized mean) reaches 0.90, outperforming both MEDIC and HomeDOCTOR baselines.

D. DISCUSSION

Our model was compared with two representative baselines widely discussed in medical NLP research: HomeDOCTOR (a RAG-based Slovenian medical assistant) and MEDIC (OCR + GPT pipeline for drug extraction). Unlike these systems, which were originally designed for Slovenian or OCR-based workflows, our model operates over a unified Franco-Moroccan corpus and applies multilingual embeddings with explicit metadata normalization. Quantitative results (Tables 4, 5 and 6) show that our system achieves a 5–8% improvement across BERTScore, Faithfulness, and Groundedness, particularly due to stricter post-generation verification and structured metadata handling. We also conducted additional experiments to evaluate generalizability across parameters, datasets, and embedding models.

Across all metrics, our multilingual RAG system outperforms HomeDOCTOR and MEDIC when evaluated on the same FR/MA corpus. Improvements are especially notable in groundedness and faithfulness, reflecting the benefit of **metadata normalization**, **multilingual embeddings**, and **NLI-based post-verification**. These gains demonstrate the importance of adapting medical RAG systems to multilingual, cross-country medication ecosystems rather than reusing pipelines built for monolingual national health systems. Compared to HomeDOCTOR, our system extends beyond national contextualization by addressing multilingual counseling. In contrast with MEDIC, which focuses on OCR-based extraction and GPT generation, our approach grounds responses in a structured and curated drug knowledge base, reducing hallucinations.

Several methodological and technical factors explain this superiority. First, the integration of curated Moroccan and French drug datasets, homogenized into a unified structure, ensures broader lexical and pharmacological coverage. Unlike MEDIC, which relies on OCR-based extraction and suffers from noise and accuracy loss, our pipeline operates on normalized datasets, minimizing extraction and generation

errors. Second, the retrieval module, built on FAISS and multilingual embeddings, guarantees higher contextual relevance than HomeDOCTOR, whose Slovenian-trained RAG model was not specifically adapted to Franco-Moroccan medical language and prescription practices.

At the generation stage, our approach achieves a better balance between semantic accuracy (BERTScore) and factual fidelity. MEDIC, although useful for OCR-driven scenarios, exhibits a tendency to generate partially ungrounded responses, reflecting a weakness in evidence attribution. HomeDOCTOR delivers competitive scores but remains limited by its training focus on European datasets, which hinders adaptation to local contexts (terminology, brand names, and cultural prescribing patterns).

These findings demonstrate that the originality of our system does not lie solely in incremental metric improvements but in its unique ability to adapt RAG-based medical chatbots to multilingual and multicultural environments. By anchoring the generative process in context-specific, curated datasets, our system addresses the pressing challenges of factual reliability, transparency, and local relevance.

This work advances the field of medical LLMs by demonstrating that retrieval-augmented, multilingual, and contextually localized architectures can outperform general-purpose LLMs and existing RAG systems. By grounding responses in curated drug repositories and extending coverage to francophone Africa, the proposed system reduces hallucinations, increases transparency, and achieves higher semantic fidelity. This not only provides a methodological contribution, by showing how to systematically improve faithfulness in medical chatbots, but also fills a critical gap in healthcare AI for underrepresented linguistic and regional contexts.

E. LIMITATIONS

It is important to acknowledge the limitations inherent in this evaluation. First, while sophisticated, automated metrics like BERTScore and faithfulness/groundedness scores may not perfectly capture all aspects of response quality, such as clinical nuance, safety implications of omissions, or overall helpfulness. Second, BERTScore is sensitive to the quality and style of the reference answers used.

Moreover, the current evaluation relies solely on automated metrics. Incorporating structured human evaluation by medical professionals, as outlined in the implementation chapter's strategy, would provide essential qualitative insights into accuracy, safety, and understandability, complementing the quantitative scores. The performance observed is specific to the test set used, which may not cover all possible query types or edge cases.

Although our evaluation demonstrates measurable improvements, the system has not yet been clinically validated, and future work will integrate expert reviews to ensure medical reliability. Furthermore, extending the pipeline to Arabic will require addressing morphological complexity and resource scarcity.

Our dataset does not currently include detailed demographic metadata (age, gender) per medication query; thus we did not stratify results by subgroup. We recognize the importance of such analysis and list it as a next step: integrating demographic labels, logging anonymized user attributes (with consent), and running subgroup performance audits to detect potential biases.

The methodological novelty of this study lies in: first, constructing and leveraging a large Franco-Moroccan medication knowledge base (13,600 records) for multilingual retrieval. Second, integrating an NLI-based evidence judge to quantify faithfulness and groundedness automatically. Third, providing a compact, domain-specific benchmark of 100 curated queries reflecting real self-medication scenarios. Practically, the system addresses a significant service gap for franco-phone populations, especially in Africa, by improving factual reliability (+5% in faithfulness/groundedness) and lowering unsafe hallucinations by 40% relative to non-grounded baselines.

VI. CONCLUSION

This work addressed the challenge of developing a reliable multilingual RAG-based framework for medication counseling, specifically adapted to the French and Moroccan healthcare contexts.

In this work, we outlined the design, development, and implementation of a system containing data preprocessing, semantic embedding using Sentence Transformers, efficient vector indexing retrieval using FAISS, and a response generation process with LLM conditioned on two contexts: the user query and the FAISS-retrieved context.

Hence, we developed and evaluated a multilingual RAG-based medical chatbot capable of providing reliable medication counseling across French and Moroccan datasets. By integrating curated repositories and optimizing retrieval-generation coupling, our system outperformed recent baselines (HomeDOCTOR, MEDIC) with a +5% margin in faithfulness and groundedness. These findings underscore the importance of grounding generative AI in structured, contextually adapted knowledge sources, especially in healthcare applications where misinformation carries high risks.

This paper highlights the increasing need for trustworthy and available medical information, by creating, deploying and evaluating an intelligent medical chatbot focused on medication-based inquiries. Recognizing the limitations of ungrounded generative medical chatbot applications, as well as the risks associated with ungrounded generative models in sensitive healthcare applications, the proposed method utilized the Retrieval-Augmented Generation architecture to ensure that the chatbot's responses were grounded in relevant and factual information, and decreased the potential harm associated with using LLMs to provide reliable medical information.

Based on the implementation experience and evaluation findings, several key directions for future work emerge to enhance the chatbot's capabilities, reliability, and usability. First, enhancing Faithfulness and Groundedness, which is the most critical area for improvement. Second, using advanced Prompting Strategies, such as Chain-of-Thought reasoning, explicit instructions to cite sources or few-shot examples demonstrating faithfulness, to better constrain the LLM's output to the provided context. Third, improving Retrieval Relevance to increase the precision and recall of the retrieval step.

Furthermore, while our evaluation focused on automated metrics, future work will involve clinical validation with healthcare professionals, and reinforcement learning strategies to further improve factual reliability. Ethical and regulatory considerations such as GDPR compliance, bias mitigation, and transparent attribution will also remain at the forefront. Ultimately, this research provides a step toward safe, multilingual, and culturally adapted medical chatbots, bridging a critical gap in access to trustworthy digital health support.

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